



Experimental geographies

Angela Last*

Department of Geography, University College London

Abstract

The proliferation of the term ‘experimental’ in human geography has given rise to the question of how geographers experiment. Given the range of different examples – from explorations of sensory methods to attempts at transforming the role of publics in decision-making – it becomes clear that one cannot talk about a unified experimental geographical approach. While projects share common themes such as challenging methodological limitations or wishing to play a more active part in the ‘production of space’, they also show fundamental differences in their attitude towards knowledge-making and intervention in the world. A starting point for further research and debate, rather than a comprehensive survey, this article outlines themes, differences and productive tensions within the discourse, and highlights the need to examine the wider politics that experimental approaches are embedded in.

Experimental Geographies

The intrinsic complexity of living systems ... does not impose a dramatic limit on any possibility of experimentation. What it imposes is the necessity for an intelligent experimentation, which assumes the risky responsibility of asking relevant questions. Isabelle Stengers (1997, p. 16)

A recent influx of papers, articles, conferences, networks and exhibitions aligning the word ‘experimental’ with ‘geography’ has given rise to a persistently growing discourse around what these ‘experimental geographies’ are or should be about. As *Powell and Vasudevan* [<http://www.envplan.com/epa/editorials/a3927.pdf>] note: ‘the vocabulary of experiment has never been more common in geographical discourse’ (2007, p. 1790). Yet what counts as a ‘geographical experiment’ or ‘experimental geography’? It could be argued that geographical, or other social scientific, research is by necessity experimental, as researchers construct an ‘apparatus’ of theoretical and methodological framing in order to produce particular results (see also Gross and Krohn 2005, p. 79). This is, of course, not the context in which ‘experimental geographies’ are invoked. Neither is it usually the scientific approach of physical geography or the ‘classic’ experimental social science (which incidentally also seems to be experiencing a renaissance, with related centres (re-)opening across universities in the UK and the US). In fact, many ‘experimental’ geographers, echoing concerns from sociologists such as John Law (2004, p. 151; see also Brown 2012, pp. 72–73; Lury and Wakeford 2012; Mann et al. 2011), actively seek to distance themselves from such ‘reductive’ forms of testing.

Geographer Doreen Massey calling for experiments to hook onto ‘cracks in the system’ in order to bring about ideological change at a recent *Interdependence Day* [[link](#)] event in London (Massey 2012a). [<http://www8.open.ac.uk/researchcentres/osrc/events/doreen-massey-4-new-maps-for-an-island-planet>]

The practices of thinking and doing that are embraced under the banner of experimentation do not comprise a unified body of work. In fact, what this article seeks to show is

that the definition, or rather, definitions, of 'experimental geography' are still open to contestation. Currently, 'experimental geographies' include anything from embedding new forms of citizen involvement in institutional processes to questioning disciplinary boundaries through the use of poetry in academic writing. 'Experimental geographies' also do not merely include work that calls itself experimental, but also the work claimed as ancestor or ally. Again, a diversity of work has become identified as such. Rather than give a comprehensive overview of the field, this article seeks to highlight some of the current tensions around experimentation, which arise not only from geography-internal struggles, but from links with debates across the social sciences and humanities. It especially seeks to draw attention to the differences between strands of experimentation and to put them into closer dialogue with each other beyond already existing overlaps. Faced with the impossibility to sketch out all of the different approaches and aims observed in contemporary practices of experimental research, this article will focus on teasing out tensions between competing understandings of experimentation reflected in geographical debates, such as the idea of experiments as means to an end and experiments as ends in themselves, the association of experimentation with practices or representation and knowledge-making and possible politics within the 'experimental society' (Krohn and Weyer 1994; see also Beck 1992). It will also address the risks involved in this kind of research, since there is hardly an experiment without risk, although it has been argued that experimentation is becoming increasingly risk-averse (Badiou and Truong 2012).

To prevent confusion about the proliferation of the term 'experimental geography' in spaces outside the discipline, it should be mentioned that not all experimental geography is undertaken by geographers: there are many examples of 'non-geographers' (e.g. architects, sociologist, anthropologists, artists, policy-makers) probing with geographical thinking as well as the imaginations and practices surrounding it. While not always constituting 'geographical research' in the narrow disciplinary sense, these engagements, ranging from art practice to governance, offer valuable ideas and prompts for geographers to re-think established ways of 'doing geography'. Doreen Massey addresses this reciprocal relationship when discussing how her concept of 'power geometries' has formed the basis of experiments with alternative forms of democratic decision-making in Venezuela and, in return, has influenced her thinking about experimentation (Massey 2008, 2012a,b). Often, such exchanges of ideas and practices have led, and continue to lead, to collaborations and have certainly influenced conceptualisations of 'experimental geography'. As experimentation is taking place under many different umbrellas, I also need to point out that some this work has already been documented in other reviews, such as art and geography (Hawkins 2011; *this journal*) [<http://onlinelibrary.wiley.com/doi/10.1111/j.1749-8198.2011.00429.x/abstract>], qualitative methods (Davies and Dwyer 2007, 2008, 2010) or cultural geography and representation (Lorimer 2005, 2007, 2008a). Readers might also want to have a look at (or participate in) the discussions on dedicated blogs, which so far include the *Experimental Research Network* [<http://experimentalnetwork.org>], *Experimental Geographies* [<http://experimentalgeographies.wordpress.com>], *Experimental Geographies in Practice* [<http://merlepatchett.wordpress.com>] or *spacesof[aesthetic]experimentation* [<http://www.spacesofexperimentation.net>]. This article will be in conversation with these and similar forums, as well as my own experiences with experimental public engagements (Last 2012a,b).

Enlivening Research and Reconfiguring Relations

How can geographers experiment? And why would they want to? To date, definitions range from the more scientific interpretation of the experiment as a testing of theories

through a carefully crafted and monitored set-up, albeit with room for the unexpected, to the less formalised 'experiment as a trial or a venture into the unknown' (Gross 2010, p. 4; Gross and Hoffmann-Riem 2005). What most geographers seem to agree on is that experimentation in their discipline should push the limitations of current conventions of representation and knowledge-making. There is a desire to move away from what is considered 'safe', orderly and established, whether it is by searching for methods that meet the imperatives of new theories, existing complexities or desired accessibility. This drive is often connected to the wish to take geographical knowledge beyond the prescribed environments and to bring it into dialogue with new disciplines, spaces and audiences (Massey 2008; Pratt and Johnson 2009; Thrift 1991). As such definitions do not necessarily make research 'experimental', some geographers have started asking whether the majority of this kind of work is actually 'experimental' and not just a strategic re-branding (e.g. for the funding of new research and/or claims to 'new knowledge') of all sorts of creative, critical or 'radical' engagements that have been taking place within and around the discipline for a long time. In fact, some experimenters have found, on some occasions, that their method yielded unexpectedly familiar outputs or that new kinds of interventions leave situations unchanged. The boundaries of the field seem hard to determine (and cases have been made why they would or wouldn't need to be). The authors of the recently established and primarily media/arts based *Experimental Research Network* (<http://experimentalnetwork.org/>) website make the argument that 'traditional research methods can be used creatively' and situate them within experimental research by including 'people who are using creative, innovative, novel or risky research practices in their work' (Gallagher and Prior 2010). Not all geographers agree on such a definition, and particular those actively involved in 'knowledge controversies' have argued for a stronger definition of experimentation and experimental risk in our field.

Other sources name interdisciplinarity as a key feature of experimental research, as experimentation is frequently prompted by the perception of discipline-specific methods as limited (see Davies 2011; Donaldson et al. 2010). The position that interdisciplinarity creates favourable conditions for generating new knowledge and practices by default has been critiqued on the basis that such research gives rise to a multiplicity of possible relations, not all of them conducive to innovation (Barry et al. 2008). For example, some disciplines share significant theoretical and methodological overlaps while others are separated by extreme contrasts in outlook, thus making different demands on the researcher in terms of producing synthesis. Still, the negotiation of differences between fields keeps being regarded as a powerful generator not only of novelty, but a useful way of seeing one's familiar approaches anew (Driver et al. 2002, p. 8).

Reflexivity also underlies the most unifying theme between different approaches: 'experimental geography' as the recognition of the geographer's implication in the construction of spatio-temporal practices and interrelations – and their deliberate amplification and mobilisation. Geographer and artist Trevor Paglen, a prominent advocate of 'experimental geographies', goes as far as proposing that experimentation is merely an acknowledgement and assertion of what geographers are doing all the time anyway: 'geographers don't just study geography, they create geographies' (Paglen 2008, p. 31). As he continues to explain:

Experimental geography means practices that take on the production of space in a self-reflexive way, practices that recognise that cultural production and the production of space cannot be separated from each another, and that cultural and intellectual production is a spatial practice. Moreover, experimental geography means not only seeing the production of space as an

ontological condition, but actively experimenting with the production of space as an integral part of one's own practice. (Paglen 2008, p. 31)

Part of his experimentation has involved art practice, such as the involvement in an exhibition entitled 'Experimental Geography (Thompson 2008) (link).

Video: [<http://www.youtube.com/watch?v=w7k5dtb741k>]

Experimental Geography Panel Discussion featuring exhibition curator Nato Thompson, artists Lize Mogel and Trevor Paglen, and geographer David Harvey.

Artist Iain Kerr, who also participated in this exhibition, suggests that 'to act/research is to be involved in change – experimental change. We need to recognise that acts of knowing are forms of change' (Kerr 2008, p. 65). This active, participatory experimentation is taking on manifold forms, which, as Gail Davies observes, are seen 'less as what can be known through precisely controlled conditions, and more about creative forms of world-making' (Davies 2011; see also Hinchliffe et al. 2005, p. 656; Hacking 1983, p. 31). Gabrys and Yusoff further suggest that 'experimental geography' is semi-consciously taking place across the world through the 'collective experiment of climate change (2012, p. 18). Following this interpretation, 'experimental geography' in the disciplinary sense would consist of a more conscious mode of intervention, or a project to make others conscious of their 'world-making'.

The active participation in 'world-making' is reflected in the desire of many geographers to move beyond 'mere critique' and to affect spaces and relations of concern through other means, which are hoped to be more effective in reaching relevant audiences (Latham 2003, p. 2012; see also Latour 2004). The search for alternative research practices or representation is often guided by the wish to align the dissemination of one's findings more closely with the ethical or aesthetic imperatives of research subjects. Among the questions that geographers have posed one finds: Do established geographical practices lead to a 'middle-classing' of social thought (Thrift 1991, p. 146)? How can we include the nonhuman in our practices and analyses (Hinchliffe et al. 2005)? How can we engage with the precognitive, with emotion (see Anderson and Harrison 2010)? Should concepts be followed formally in writing (Massey 1997) and certain impressions be rendered as poetry (Lorimer 2008b)? Should writing on experimentation result in experimental writing? These kinds of questions emphasise the intertwining of aesthetics, ethics and ways of knowing and representing; they ask what can be known and represented, and what kinds of options are available for engaging with the potentially unknowable and unrepresentable through experimentation.

The capacity of experiments to surprise becomes strong attractor in the play between the 'known' and the unknown (Massey 2005, pp. 29, 111; McCormack 2010, p. 217; Pinder 2005, p. 390). As various historians of science have shown, surprises are likely to occur, no matter how controlled the set-up (Rheinberger 1992, p. 307; see also Ronell 2003, pp. 563, 568). Avital Ronell's description of experimentation gives an indication of the risk, but also the appeal for those willing to abandon the known: '[t]he experimenter must give up any secure anchoring in a homeland, allow herself to be directed by an accidental current rather than aiming for a pre-established goal' (Ronell 2003, p. 568). Rather than serving as a deterrent, the prospect of co-producing the unknown seems to be fuelling the drive towards experimentation. Affecting future events through setting up conditions that give rise to indeterminate encounters is one way of engaging with the unknown or unknowable (Figure 1).

Explicitly pushing this point, Adeola Enigbokan's project 'Foolish Journeys' involves participants in imagining new methods for experiencing and 'archiving' the city through



Fig. 1. Image from Adeola Enigbokan's *Foolish Journeys*. Image author: Adeola Enigbokan.

practices such as tarot readings and the making of (im)possible maps (2011) [<http://archivingthecity.com/2011/04/23/foolish-journeys-documentation/>]. Such work not only addresses issues of how to engage with or even shape the unknown, but further suggests an unknowable transformation of the experimenter herself (Ronell 2003, p. 569), an aspect that plays a fundamental role in terms of the outcomes of experimentation and tends to be underexamined: who or what is really being transformed, and to what ends? This emphasis on the unknown is not only raising questions about hard-to-rationalise experience but also about research ethics, an issue that will be addressed in the course of this article. Further, work pursuing the goal of surprise in an extreme form inadvertently, and sometimes deliberately, goes against the grain of current research regulation, which, while seeking to promote innovation, tends to favour it along predictable lines and does not tolerate failure. Here, difficult sites of negotiation emerge: how much risk can be taken?

The fixation on unknowability and surprise is complemented by questions of experimental rigour: what makes an experiment, what are its criteria for success or failure – or do experiments need to conform to such stereotypical values? Again, answers differ significantly. It seems useful to start by contrasting of two prominent strands of discourse around experimentation, one commonly associated with cultural geography, but not limited to this field, and the other emerging from geographical engagements with science studies. Although these discourses share common theoretical influences, for instance, in the work of theorists such as Hans-Jörg Rheinberger, Isabelle Stengers, Bruno Latour and Avital Ronell, they seem preoccupied with different aspects of experimentation.

Representational Interventions, Inventive Representations

The first area of experimentation I would like to introduce spans a variety of work in human geography with an interest in performativity, sensation and representation. Judging by recent titles (and content) of articles, blogs and conference sessions, this area seems to have become almost synonymous with 'experimental geography', despite contestations of its experimental nature. I will deliberately force some often diametrically opposed work together to produce this 'area', for the reason that different groups have been making claims to 'ancestry' over very much the same methods and strategies, from

'sensational Marxists' [<http://meridian.aag.org/callforpapers/program/SessionDetail.cfm?SessionID=12233>] to 'non-representational theorists' (Thrift 2004; Thrift and Dewsbury 2000). It is particularly in this discourse that the range of 'classic' geographical methods and outputs, such as taking fieldnotes, doing interviews and construction written critiques, is perceived as perpetuating the same kinds of exclusions and as too limited to make a difference in the world, as well as for fellow academics and/or for oneself (Crang 2003; Loftus 2012, p. 136; Lorimer 2005, p. 83). The perceived lack of methodological and stylistic richness has been contrasted with the amount of theoretical experimentation taking place within the discipline (Thrift 2000). It could be argued that this perception of lack is being extended to all of academia, with the exception of the arts, since through comparing the different approaches, a common tendency to privilege artistic registers becomes apparent (incidentally, a similar debate around representation and disciplinary boundaries is taking place in art, which I will draw on in the final part of this article). The utilisation of these registers does not only appear to be prompted by an association of art with greater immediacy, affectivity or freedom of expression, but also by a desire to actively participate in the creation of new geographical imaginaries. Moreover, questions have been raised about the omission of art-specific methods from the geographical repertoire (Wylie 2010, p. 101). This does not necessarily mean that geographers partake in producing art, but follow research paths prevalent in the arts, such as practice based research, which in many cases involves the creative use of media such as photography, film and sound recordings. Over a decade of such practice based geographical research can, for instance, be found in the 'Cultural Geographies in Practice' section of the Cultural Geographies Journal.

Whether located within the arts or other disciplines, methods regarded or reinterpreted as non-traditional geography become a way of interrogating and transcending disciplinary boundaries. DeLyser and Rogers, for instance, regard performative methodologies as 'a situated, embodied engagement with *how* and *why* we as scholars choose to undertake specific research in specific ways' (2010, p. 188). Geographers who are sympathetic to this form of questioning have answered the 'why' in a diversity of ways, from seeking 'artistic licence' for its own sake (Wylie 2010, p. 111) to wishing to engage researched communities for ethical reasons (Pratt 2000). In the struggle with representing the complexities and openness of their field of research, and extending the reach of one's research to new communities, finding new ways of 'writing' becomes an imperative (Cook et al. 2007). As David Pinder writes, this search should take place 'not for its own sake but in recognition of the politics as well as poetics of representation' (2005, p. 402). What has been 'written out' and how can we rewrite it (Greenhough 2010, p. 51)? Thus, question of representation could be regarded as the most central issue of this strand.

Attitudes to representation determine media, styles as well as the audiences of experimentation. For many geographers, textual experiments remain a first point of call. Here, perceived scholarly dryness and/or inability to convey important aspects of research that exceed acceptable frames of representation are substituted or supplemented with creative writing ranging from poetry and collage to fiction. The edited collection 'Patterned Ground' (Harrison et al. 2004; for a related project see Pile and Thrift 2000) plays with the form of the dictionary, deliberately diverging from it, particularly the seemingly 'neutral' stance of expertise. To fulfil its project to 'unsettle prevalent understandings of the relationship between nature and culture', the emphasis is placed on impassioned engagement and fluidity and production of meaning (Harrison et al. 2004, p. 9). In a similar questioning of authoritative narrative, Caitlin DeSilvey's (2012) story of a Cornish harbour aims to unsettle ideas of stasis in conservation projects by stressing change across

time. As part of the proposed intervention, poetry and fiction are presented as forms which enable a more subjective, sensual engagement with the subject.

Geographical traditions of landscape writing constitute a frequent starting point for textual experimentation. The sensory approach to landscape of humanist geographers such as Yi Fu Tuan (1977) and J Douglas Porteous (1990) presents a starting point for many experimental 'landscape writers', not necessarily as something to be emulated, but something to build on or diverge from (Lorimer 2008b, p. 182; Macpherson 2005, pp. 96, 99). Recent work in this direction has emphasised personal experience and, through the parallel running of different genres of narrative tracks, has played with theories and (non)disciplinary practices around landscape. Lorimer and Wylie's 'Loop' piece (2010), published in a journal for performance studies, merges poetry, story-telling and academic writing to relay the authors' walking journey. More so than 'Loop', Shiloh Krupar's narrative stresses the conflict of author investment/emotion and academic enquiry. Her 'ethno fable' runs alongside an 'academic and personal subsurface guide', the latter taking on the form of excessive footnotes (2007, p. 194). Krupar explains her reasons for using what she calls a 'performative representational strategy' as follows:

- (1) to produce a certain affect of curiosity, concern, and outrage at the staging of nature spectacle on militarized sites by organizations that continue to produce and profit from deadly wastes;
- (2) to show the various discourses and representations, figures, material practices, institutions, and personal experiences of the author that have constellated in and around this site-based study;
- (3) to display two texts; one being the performance script that displays some of the rhetorical contrivances of the Rocky Mountain Wildlife Refuge nature spectacle, and the other, a supplementary text that attempts to contaminate the clean surface of the site and its staged unchanging wilderness, interjecting academic substrata and dumping a personal landfill of mythic histories, alternative landscape taxonomy, documentary photography, and animal avatars, or, subsurface tour guides (2007, p. 195).

Other work has more explicitly stressed the need to move beyond the merely 'affected' and to move on to action or the imagination of alternatives. David Harvey, ending his book 'Spaces of Hope' with the science fiction story 'Edilia', calls for the need to generate further stories and possibilities (2000, p. 258).

Experiments with visual representation have taken place mainly in the form of collaborations between artists and geographers, although more and more artistic projects are undertaken by geographers themselves. Collaborations tend to define their experimental nature through the exploration of disciplinary tensions between the two groups involved. Examples of such collaborations include the projects 'Landing' (Driver et al. 2002), 'Fashioning Diaspora Space' (Figure 2) [<http://www.vam.ac.uk/content/articles/f/fashioning-diaspora-space-introduction/>] 'Describe a Landscape' (Tolia-Kelly 2010) and 'MiceSpace' [<http://www.micespace.org/>], and the artist residency at the UCL Urban Laboratory [<http://www.ucl.ac.uk/urbanlab/en2/index.php?page=1.2.7>]. Looking and listening to the various reports on such collaborations, they seem to make clear that disciplinary differences may end up being complicated rather than offer safe starting-points, as project participants discover significant similarities between approaches (Driver et al. 2002, p. 7). During 'Landing', geographers 'interested in more imaginative or inventive research methods' found themselves confronted with 'artists committed to rigorous empirical research' (Driver et al. 2002, p. 8).

Geography-internal experiments, while united by a dissatisfaction with geography's attention to the visual, appear to originate from two different directions (of course, there are overlaps). The most prominent direction draws attention to geography's visual history,

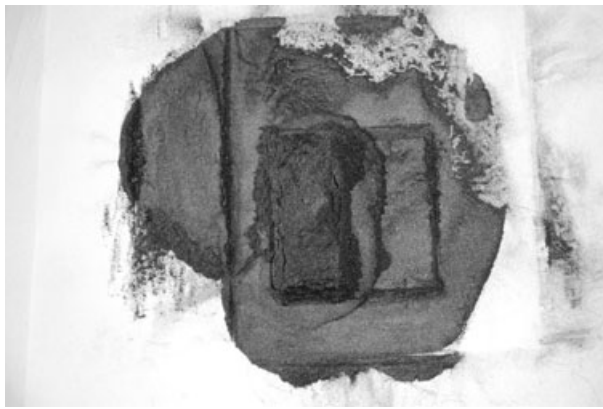


Fig. 2. Image from MiceSpace collaboration between artist Helen Scalway and geographer Gail Davies. The projects grapples with rendering the spatial, ethical and political complexities of global laboratory mice geographies. *Image author: Helen Scalway.*

such as the images found in the archives of geographical societies or popular culture (Driver 2003, 2012; Yusoff 2007). Visualisations such as photography, film, sketches and maps are regarded as geographical techniques whose content and means of production is subject to critical contestation (Rogoff 2000). On the other hand, there are geographers who see visual media as a novel introduction to the discipline, or at the 'expressive' registers of it (Crang 2003, p. 501). By re-interpreting geographic visuals or visual tools, the former are trying to create alternatives to archetypal representations such as the heroic explorer or 'factual' maps. Kathryn Yusoff's 'remixing' of an Action Man Antarctica special, turned into 'Action Man – Antarctic Explorer' into 'Action Man – Polar Inertia', asks us to consider 'non-productive landscape encounters' against a background of territorial conquest (2007, p. 211). Mapping equally becomes a target of critique as a process which 'depends upon scientific experimentation and a representational view of the world' (Kitchin et al. 2009, p. 8). After the contestations of psycho-geographers and authors such as Cosgrove (1999), Bunge (1969) and Harvey (1969), 'critical cartographers' have turned to artistic or participatory experiments in map-making that emphasise the subjective, the provisional, the excluded and the unforeseen (Crampton 2009; Crampton and Krygier 2006; Kitchin et al. 2009). These have taken a variety of formats, including graphic novel-like 'propositions' (Krygier and Wood 2009) [http://academics.skidmore.edu/blogs/ar-357-f10-kikedai/files/2010/09/ce_n_est_pas_le_monde.pdf], 'nonhuman maps' (Harris and Hazen 2006) and 'evolving' open source online projects (Williamson and Connolly 2009). Contrarily to Yusoff's desire to de-emphasise productivity, 'critical cartography' appears to re-emphasise it, by calling for map-making to be more actively in the service of producing worlds for example, improving social relations between different communities (Kitchin et al. 2009, p. 16; see Bunge 1969).

Even more so, audio-visual methods, privileging immersive, sensory experience, lay claim to expanding and enriching disciplinary practices. It has been argued that, in the sphere of non-textual methods, there has been a dominant focus on the visual, which has, so far, hindered an engagement with other kinds of sensory registers (Edensor 2007, p. 219; see also Crang 2003, 2010; Rose 2003). Dubbed by some researchers as the 'new empiricism of sensation' (Clough and Halley 2007), experimentation becomes a method to find ways of embedding non-traditional sensorial range(s) and contesting a reliance on

vision and prose. These include attempts to address 'pre-reflective' events by immersing oneself in particular environments and practices from dancing to diving (McCormack 2003; Merchant 2011) [<http://theoryculturesociety.blogspot.co.uk/2011/06/interview-with-stephanie-merchant.html>]. Although excessively documented through photography, urban exploration and other forms of engagements with (dys)functional and ruined spaces also lay claim to non-visual experimentation (Battista et al. 2005; Edensor 2007). As Tim Edensor proposes, experience of smell, sound, temperature and texture is put in the service of 'multiply[ing] the meanings and experiences of place, open[ing] out an awareness of sensory alterity and banish[ing] illusions of essence and fixity' (Edensor 2007, p. 230).

Video: <http://vimeo.com/17235930#>

The short film 'Jute', an experiment to 'catalyse[...] sight, sound, thought and action' (Rosa et al. 2010). Showcased on the 'Experimental Research Network' website.

The emphasis on personal experience of much sensory research seems to suggest that it is merely invested in reflecting on personal experience, however, there are many examples of projects which focus on reaching new audiences with academic research and building experimental practices which last beyond the duration of the experiment. Geraldine J. Pratt's use of 'testimonial theatre' intended to transform relations between Filipino domestic workers and their Canadian employers. Reporting on the unexpected ways boundaries between theatre and 'real life' became redrawn during the play – with audience members uncertain whether they were witnessing actual 'victims' or actors, or whether they could intervene in the play by asking the actors questions, Pratt suggests that they lead to new configurations between people after the play, for instance, leading to the formation of citizen initiatives and other forms of learning (2009, p. 132). Other projects have attempted to promote civic dialogue by defamiliarising everyday (as well as iconic geographical) practices such as walking (Ramsden 2011).

Video: <http://vimeo.com/1569921>

A preview for 'Nanay – A Testimonial Play'

Despite the apparent drive towards creating or re-naming existing research that contests disciplinary modes of representation as 'experimental geographies', discussions of experimentation in this strand rarely seem to go beyond celebration of characteristics such as novelty or openness or the refutation of experimentation as narrowly instrumental (some 'experimental geographers' actively refuse 'useful' or 'rational' outcomes Rosa et al. 2010). Exceptions include Derek McCormack's attempt to forge links between a focus on experience and 'affirmative' experimentation (2011). Taking inspiration from the work of William James and John Dewey, who argue for experimentation as a generative experience that can take place anywhere, McCormack proposes to view experimentation as a means of negotiating critical thinking and affective response (2011, p. 212). Other discussions of experimentation have taken place via science studies, which has also given rise to a second form of 'experimental geography'.

How to know and not know

The second direction of experimental geography, although also interested in representation and organised around similar themes such as surprise, interdisciplinarity and openness, places a stronger emphasis on examining how 'new' knowledge is produced. Taking inspiration from the work of philosophers and sociologists of science such as Galison (1987), Hacking (1983), Haraway (2008), Latour (Latour and Woolgar 1979), Rheinberger

(e.g. 1992), Shapin and Schaffner (1985), Stengers (1997), but also theorists such as Ronnell (2005), geographers ask how experiments are organised to generate 'differences', but also how these differences depend on – and create – systems that recognise and validate outcomes. The critical engagement with the history, rhetoric and wider politics of experimentation forms part of this inquiry, in order to build an idea of what experimentation is and does. Geographers of this direction seem to be joining together 'geographies of experiment' (the identification and analysis of 'experiments' they come across (e.g. *Driesen and Lorimer* 2011) [<http://wildexperiments.com/>] and 'geographical experimentation' (the staging of experimental interventions to generate new knowledge and practices).

In terms of organisation, it is the spectrum between the obvious limitation of experiments – their anchoring in familiar ways of thinking and practices – and their uncanny potential for unexpected events that attracts researchers, whether it is in the scouting out of potential for applications of new theories or the merging of existing methods with new influences.

This direction is also less hostile to traditional forms of representation, referring to studies such as those of Hans-Jörg Rheinberger who finds 'individual artistic potential' in the zone between event, fieldnotes and the construction of the argument for an written piece of research (2003, p. 626; see also Barad 2007, p. 55 on theorising as a material intervention). By contrast, sensory forms of experimentation are more likely to be viewed with suspicion for their associations with seduction and the engineering of complicity or uncritical involvement (Marres 2009, pp. 118, 123; see also Bougnoux 2006, p. 77).

The reason why this area has been less associated with 'experimental geography', apart from its own potential theoretically motivated objections, is the nature of its projects. While there are many geographers writing about experimentation and knowledge-making, fewer projects seem to be undertaken. The kinds of projects that are undertaken, however, tend to involve larger numbers of people and are often interdisciplinary in nature. Another factor is the overlap of this kind of research with similar projects in sociology and science and technology studies (STS). So far, most of this work appears to be centred around environmental and technological controversies. Here, geographers and other social scientists have identified unhelpful practices of knowledge-making, which they seek to transform. This is often described as an 'experimental reassembling' of existing practices (Hemment et al. 2011). The project 'Understanding Environmental Knowledge Controversies' [<http://knowledge-controversies.ouce.ox.ac.uk/>], for instance, set out to reposition scientific expertise in flood management by integrating the competencies of 'non-scientist publics' (Lane et al. 2011). The main method for achieving this involved bringing together scientists, social scientists and what tends to be classified as the 'victims' of flooding – the affected 'public' – and by discussing solutions that would normally not have been deemed acceptable under the previous model of flood prevention (Lane et al. 2011, p. 31). Projects such as 'Deliberative Mapping' (Davies et al. 2003) or 'Governing at the Nanoscale' (Kearnes et al. 2006) are characterised by similar aims and structures. The 'Deliberative Mapping' website [link: <http://www.deliberative-mapping.org/about.htm>] sums the structure up as follows: examine, develop, test.

In practice, this form of experimentation is likely to look far less linear. This became apparent during my own experimental engagement, 'Mutable Matter', an experimental public engagement with nanotechnology which sought to demonstrate how representations of material agency affect political agency of different publics (Last 2012). The stages alternated continuously: testing informed a re-examination or re-development, and sometimes, a development took place prior to examination. This does not imply that the stages do not accurately represent the process – after all, they remain the underlying logic,

and the capacity of these experiments to capacity to succeed or fail. Despite making a case for a degree of playfulness as part of experimentation and the kind of radical openness that emerges during interaction with the world (Galison, 1987, p. 278; see also Woodyer 2012) – certainly they had a substantial role in my own projects – the second strand is making a strong argument against the notion of too open-ended experimentation. Experimenters see a danger of not arriving at anything at all. As STS scholar Sheila Jasanoff suggests: ‘in the absence of paradigmatic agreement about how to interpret experimental results, you are liable to get an infinite deconstruction of the results’ (Jasanoff 2006). If one does not agree on the experimental parameters of the experiment, how can one gauge the attempted contribution (see Rheinberger 1992, p. 324)? To arrive at a conclusion, experiments may not have to have a fixed hypothesis to be tested, but can be ‘experiential’, in the sense that ‘we try something out through experience’, which equally allows us to determine success or failure (Lane et al. 2011, p. 24). For example, the success of the flood management experiment was measured by the positive responses to the strategy (the feedback of the participants, the up-taking of the model by other communities facing the same threat) and the somewhat unexpected outcome of a new flood management model (how to route the rainfall) (Lane et al. 2011, pp. 25–26).

To set criteria and to ‘limit unknowns’ is also regarded as a matter of ethics. What may happen during an experiment not only causes substantial anxiety for the researcher and researched, but also for external funding bodies, ethics committees and the wider public. During the ethics consultation of ‘Mutable Matter’, the inclusion of art practice based methods (Figure 3) initially led to a refusal from a more quantitatively orientated committee. The proposed methods, and their association with indeterminable outcomes, fell outside the frame of research value and of ‘educational value’ to the participant group. Further proof of its value, and its status as an ‘innovation’ rather than ‘playground’, had to be assembled in the form of cross-referencing with other projects, pilot outcomes and a more in-depth explanation of the theory and method relationship. Ethics are also a factor why many geographers have been suspicious of the rhetoric of experimentation in relation to their work. As part of the research for this article, I made enquiries about experimental projects in other areas invested in struggles with unequal access to constructions of knowledge (e.g. in relation to minorities, development). While many analyses of non-academic projects such as citizen initiatives, amateur archiving or NGO trials

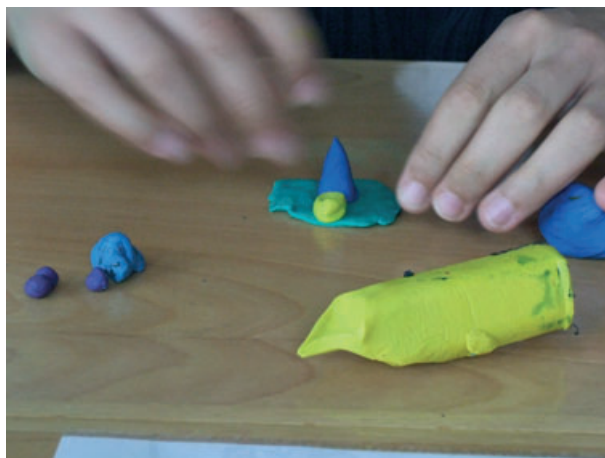


Fig. 3. Image from a Mutable Matter workshop in Leeds. *Image author: Angela Last.*

surfaced, it seemed impossible to find comparable examples of geographers actually conducting an 'experiment'. Apart from the possibility that a different vocabulary may be used for similar work, which would make such projects harder to identify, I was also pointed to the uneasy history of much of this research with experimentation. As Powell and Vasudevan (2007) note, despite geographers' enthusiasm for the complicated negotiations of power relations around experimentation, they have looked surprisingly little into the 'experimental past' of their discipline. Unsurprisingly, geographers' involvement in experimentation can be traced to exploration and to the creation and maintenance of colonies, which have become described as social laboratories (Bonneuil 2000; Driver 2004, p. 93; Stafford 1989, p. 201). Considering the critical stance that geographers have taken towards their past, a surprising amount of 'experimental research', particularly of the previous strand, takes the experimentation/exploration combination as a role-model, whether consciously or unconsciously, for instance, through its sometimes uncritical referencing of geographical stereotypes such as the (white male) heroic explorer. While such criticism has been countered with arguments for satire or re-interpretation (Battista et al. 2005), it has also been publicly dismissed by some (Garrett 2011; Garrett quoted in Barkham 2012).

Looking at geography's history of experimentation highlights the need to pay attention to the participation of such research in a wider economy. The links of experimentation not only with exploration, but exploitation, that seem so evident from geography's past, are equally present today. Such concerns extend into the current conversation around the on-going 'laborisation of life' (Callon et al. 2009), the notion that 'the experimental [is] witnessing a shift from *knowledge-site to social process*' (Corsín Jiménez and Estalella 2010; authors' italics, see also Ronell 2003). Not only has the experimental expanded out of the laboratory into the world (if it ever was contained there), but, in addition, every person seems subjected to a growing imperative to experiment (and be experimented on) at all levels of life (Agamben 2009, p. 15; Beck 1992; Ronell 2005). In recent political debates, 'citizens' have increasingly been assigned the status of potential substrate for the generation of ideas, which can translate into commercial innovation (Salmelin 2011; Thomke 2003). The drive towards the wider promotion of experimentation especially seems to sabotage the intention to use experimentation as a strategy against the instrumentalisation of research for economic ends. A recent *conference on 'experimentality'* [<http://www.lancs.ac.uk/experimentality/>] asked the uncomfortable question: 'is the experiment now too complicit with power to act as a carrier of hope?' (Lancaster University, 2010)

Video: http://www.dailymotion.com/video/xjyb2x_br-or-salmelin-le-futur-de-la-creation_tech
European Commission advisor Bror Salmelin discussing 'experimental mentality' in the context of innovation

The experimentation-exploration-exploitation link is further helpful in sensitising geographers to the question of who is in a position to experiment. Some geographers have worried that too much experimentation is centred around the notion of the privileged, heroic (and gendered) experimenter (see Davies 2010, p. 4; Kearnes 2007, p. 149; White 2010). Following Geraldine Pratt's concern about who gains in a research relationship (2000), several questions need to be asked: Who gains in an experimental relationship? Who can participate in 'making the future' (François Jacob quoted in Rheinberger 2003), in making worlds? How is agency distributed during (and beyond) the experiment? In this context, it has been debated whether only 'those in power' are able to experiment or

whether less privileged or 'passive' experimental subjects are able to 'experiment back'. In both post-colonial and science studies, the controversial case has been made that both colonial and 'nonhuman' subjects are indeed in a position to do so (Beisel Forthcoming; Latour 2000; Philips 2006; Tilley 2011). If this is the case, to what degree are they able to and how is this represented? It is especially against this growing pressure to experiment that continued exchange and self-reflexivity about personal practices and their wider implications are of the essence.

Conclusion: Experimental Encounters

This difference in emphasis has led to the situation where the two discourses could potentially talk past each other, but could also open themselves to questioning the other's approach. At the core of the potentially productive tensions appears to be not merely a disagreement on the mechanisms of the experimental or what kind of knowledge is produced and how, but a difference in the perception of 'creativity' in geography. The first strand primarily bases experimentation on the need to enliven research practices and the interface between research content and audience through creative means and representation from 'outside', grounded in an assessment of geography – and the academy – as limited, utilitarian or self-referential. Against this stands the second strand's belief in the creativity inherent in geographical/academic research through practices such as conceptualisation. First attempts soften or dissolve the presumed binary of artistic-creative and academic-utilitarian research have been made by de-romanticising and 'demystifying' artistic practice (Croll-Knight 2010) or by showing 'making as already a thinking-in-action, and conceptualisation as a practice in its own right' (Jellis 2011; see also Corsín Jiménez and Estalella 2010).

A meeting ground has been identified in debates around value of experimentation as a productive intervention. It is generally assumed that experiments have value and that this value needs to be protected. The second strand addresses the danger of weakening the experiment's creative force by setting it up in a climate of 'anything goes': if experimentation is too open, the value of experimentation is likely to be deflated, no longer providing challenge, but merely the appearance of a challenge. Loss of value potentially also occurs if all creative practices are mistaken for challenging or knowledge-generating practices. The situation becomes problematic when 'experimental' research seems to follow in the tracks of already existing theories and practices, or becomes preoccupied with learning new techniques rather than moving the field into new directions. In such cases, the infatuation with the idea of experimentation could lead to the privileging of an 'experimental aesthetic' over truly experimental methods packaged in more familiar forms. In the first strand, fears have been voiced that uncritical approaches to sensory methods could lead experimenters to fall prey to what David Buckingham calls 'naïve empiricism' (Buckingham 2009). Buckingham cautions against the strong temptation to think that 'non-conventional methods' allow a privileged access to 'parts that other methods have failed to reach' (2009, p. 635). Other authors have invited debate on whether too much enthusiasm for new methods could lead to misrepresentation or missing out on particular insights (Hitchings 2012).

The emphasis on value does not answer, however, why experimentation has become so prized. The nature of the attraction appears to lie between the general drive towards an 'experimental society' and personal desire to utilise the generation of difference created by experimentation and its potential use against appropriation by the systems under critique (Thrift 2004, pp. 40–55). Overall, conversations have identified a need for a greater reflexivity around the politics behind the new 'relationships of power' which come into

being through experimental practices (Buckingham 2009; Szerszynski 2005, p. 188). At the smaller scale, going outside established channels requires making new alliances and creating new dependencies, especially as some of the art-practice based experimental work can be rather costly in terms of training, access and materials. At the larger scale, funding and research politics have been on the agenda: while some geographers, especially of the representational strand, tend to complain about the side-lining of 'experimental' methods because of its inability to fit the 'boxes' of existing support networks, others (or the same people) have noted a biased promotion of experimentation in the increasing demand from funding bodies to use 'experimental methods' to produce 'innovative work', particularly work that creates new audiences or economically useful outcomes (Barry et al. 2008; Foster and Lorimer 2007, p. 431; Patchett 2011; see also Strathern 2006). The extent to which this happens has given rise to the concern that linking 'experimental', 'innovative' and 'research revenue' can potentially sideline more traditional, but nevertheless vital research and could lead to a disempowerment rather than an enrichment of means specific to geography.

Re-considering Badiou's suggestion that experimentation perhaps needs to be more 'risky', in this case to offer some resistance to current innovation politics, a question might be whether increased risk-taking (at least temporarily) guards against appropriation. Here, some outcomes of the debate around 'relational aesthetics' might be helpful. The essence of this debate can be summed up as follows: should art/experimentation be in the service of 'making things better', potentially only providing temporary, small scale 'band aid' solutions for problems that need to be shaken up at a larger scale, or should art/experimentation take on antagonistic features (Bishop 2012; Bourriaud 1998)? It could be argued that antagonism already somewhat informs current practices. In parts of the first strand, the imagination of geographical experimentation as located in the margins of the acceptable seems to inform the strategies such as 'useless' experimentation and sensory or ornamental excess to prevent utilisation. At times, the mere potential of experimentation for bringing forth something entirely 'non-sensical' is hoped to make some geographical engagements more resistant against absorption into the very systems it is seeking to question (see Paglen 2008, p. 32). But how useless do geographers want to (or have to) be? Within the second strand, growing disillusion with the effectiveness of experimental intervention in the governance of knowledge (Kearnes and Wynne 2007) alternates with renewed attempts to make a difference through new experiments. For Isabelle Stengers, 'risky responsibility', as mentioned in the quote opening this article, implies a close interrogation of one's methods ('no method is neutral' Stengers 1997, p. 16) and one's sensitivity to what is relevant. If we gauge this wrongly, we risk 'silencing the very thing [we are] interrogating' (op. cit). An important future direction may thus be for experimenters from both strands to remind one another what the ends of research are and whom they serve, and to ask what sort of world(s) can we hope to create – and how – under the current conditions.

Acknowledgements

The author would like to thank Gail Davies, Mireille Roddier, Uli Beisel, Adeola Enigbokan, Thomas Jellis, Divya Tolia-Kelly and two anonymous reviewers for thought-provoking comments and discussions. Thank you also to Ian Cook for his editorial guidance, to Celia Lury for the 'Inventive Methods' preview and to Harriett Hawkins for multiple conference panel organisation and for article previews.

This article benefitted from the research undertaken during my ESRC Postdoctoral Fellowship PTA-026-27-2869.

Short Biography

Angela Last is a geographer with a background in art and design. She recently completed a post-doctoral fellowship in Geography at University College London and currently teaches on the MA Art and Science at Central Saint Martins. Angela researches the relationship between representations of material agency and (human) political agency, which she pursues through 'theoretical experiments' as well as 'experimental public engagements'. The latter have included art practice based projects around new technologies and environmental risks. Her latest project examines climate change and ideology. Angela's blog Mutable Matter can be found at <http://mutablematter.wordpress.com>.

Note

* Correspondence address: Angela Last, MA Art & Science, Central Saint Martins, Archway Campus, 2 Elthorne Road, London N19 4AG, UK. E-mail: a.last@csm.arts.ac.uk.

References

- Agamben, G. (2009). *What is an apparatus? and other essays*. Stanford: Stanford University Press.
- Anderson, B. and Harrison, P. (2010). The promise of non-representational theories. In: Anderson, B. and Harrison, P. (eds) *Taking-place: non-representational theories and geography*. Aldershot: Ashgate, pp. 1–34.
- Badiou, A. and Truong, N. (2012). *In praise of love*. London: Serpent's Tail.
- Barad, K. (2007). *Meeting the universe half-way: quantum physics and the entanglement of matter and meaning*. Durham and London: Duke University Press.
- Barkham, P. (2012). The illicit art of urban exploration. The Guardian [Online] 9 April 2012. Retrieved on June 2012 from: <http://www.guardian.co.uk/uk/shortcuts/2012/apr/09/urban-exploration-student-shard-urbex>.
- Battista, K., LaBelle, B., Penner, B., Rendell, J. and Pile, S. (2005). Exploring 'An Area of Outstanding Unnatural Beauty': a treasure hunt around King's Cross, London. *Cultural Geographies* 12 (4), pp. 429–462.
- Barry, A., Born, G. and Weszkalnys, G. (2008). Logics of interdisciplinarity. *Economy and Society* 37 (1), pp. 20–49.
- Beck, U. (1992). *Risk society: towards a new modernity*. London: Sage.
- Beisel, U. (forthcoming). The blue warriors – ecology, participation and public health in malaria control experiments in Ghana. In: Geissler, W. (ed.) *Changing states of science: ethnographic and historical perspectives on government, citizenship and medical research in contemporary Africa*. Durham and London: Duke University Press.
- Bishop, C. (2012). *Artificial hells: participatory art and the politics of spectatorship*. London: Verso.
- Bonneuil, C. (2000). Development as experiment: science and state building in late colonial and postcolonial. *Osiris* 15, pp. 258–281.
- Bougnoux, D. (2006). *La crise de la représentation*. Paris: La Découverte.
- Bourriaud, N. (1998). *Relational aesthetics*. Paris: Presses Du Réel.
- Brown, S. D. (2012). Experiment – abstract experimentalism. In: Lury, C. and Wakeford, C. (eds) *Inventive methods: the happening of the social*. London: Routledge, pp. 61–75.
- Buckingham, D. (2009). 'Creative' visual methods in media research: possibilities, problems and proposals. *Media, Culture & Society* 31 (4), pp. 633–652.
- Bunge, W. (1969). *The first years of the detroit geographical expedition: a personal report*. Detroit: Society for Human Exploration.
- Callon, M., Lascoumes, P. and Barthe, Y. (2009). *Acting in an uncertain world: an essay on technical democracy*. London: MIT Press.
- Clough, P. and Halley, J. (2007). *The affective turn: theorising the social*. Durham, NC: Duke University Press.
- Cook, I., et al. (2007). 'It's more than just what it is': defetishising commodities, expanding fields, mobilising change. *Geoforum* 38 (6), pp. 1113–1126.
- Corsín Jiménez, A. and Estalella, A. (2010). Prototyping relationships: on techno-political hospitality. *ARC*, [Online]. Retrieved on June 2011 from: <http://anthropos-lab.net/studio/prototyping-relationships-on-techno-political-hospitality/>.
- Cosgrove, D. (1999). Introduction: mapping meaning. In: Cosgrove, D. (ed.) *Mappings*. London: Reaktion Books, pp. 1–23.
- Crampton, J. W. (2009). Cartography: performative, participatory, political. *Progress in Human Geography* 33 (6), pp. 840–848.
- Crampton, J. W. and Krygier, J. (2006). An introduction to critical cartography. *ACME: An International E-Journal for Critical Geographies* 4 (1), pp. 11–33. URL: <http://www.acme-journal.org/vol4/JWCJK.pdf>

- Crang, M. (2003). Qualitative methods: touchy, feely, look-see. *Progress in Human Geography* 27 (4), pp. 494–504.
- Crang, M. (2010). Visual methods and methodologies. In: DeLyser, D., Herbert, S., Aitken, S., Crang, M. and McDowell, L. (eds) *The SAGE handbook of qualitative geography*. London: Sage, pp. 208–225.
- Croll-Knight, T. (2010). Creative practice primer project: dundee transect. Experimental Research Network. [Online]. Retrieved on June 2011 from: <http://experimentalnetwork.org/creative-practice-primer-2/creative-practice-primer-undee-transect/>.
- Davies, G. (2010). Where do experiments end? *Geoforum* 41 (5), pp. 667–670.
- Davies, G. (2011). Molecular life. In: Del Casino, V., Jr, Thomas, M., Cloke, P. and Panelli, R. (eds) *A companion to social geography*. Oxford: Wiley-Blackwell, pp. 257–274.
- Davies, G. and Dwyer, C. (2007). Qualitative methods: are you enchanted or are you alienated? *Progress in Human Geography* 31 (2), pp. 257–266.
- Davies, G. and Dwyer, C. (2008). Qualitative methods II: minding the gap. *Progress in Human Geography* 2 (3), pp. 399–406.
- Davies, G. and Dwyer, C. (2010). Qualitative methods III: animating archives, artful interventions and on-line environments. *Progress in Human Geography* 43 (1), pp. 88–97.
- Davies, G., et al. (2003). Deliberative mapping: appraising options for addressing ‘the Kidney Gap’. Wellcome Trust Final Report, Grant 064492. [<http://www.deliberative-mapping.org>]
- DeSilvey, C. (2012). Making sense of transience: an anticipatory history. *Cultural Geographies* 19 (1), pp. 31–54.
- Donaldson, A., Ward, N. and Bradley, S. (2010). Mess among disciplines: interdisciplinarity in environmental research. *Environment and Planning A* 42 (7), pp. 1521–1536.
- Driessen, C. and Lorimer, J. (2011). Wild experiments. [Online]. Retrieved on June 2011 from: <http://wildexperiments.com/>.
- Driver, F. (2003). On geography as a visual discipline. *Antipode* 35 (2), pp. 212–243.
- Driver, F. (2004). Colonies. In: Harrison, S., Pile, S. and Thrift, N. (eds) *Patterned ground: entanglements of nature and culture*. London: Reaktion Books, pp. 93–94.
- Driver, F. (2012). Hidden histories made visible? Reflections on a geographical exhibition. *Transactions of the Institute of British Geographers* [online preview].
- Driver, F., Nash, C., Prendergast, K. and Swenson, I. (eds) (2002). *Landing – eight collaborative projects between artists and geographers*. London: Royal Holloway.
- Edensor, T. (2007) Sensing the ruin. *The Senses and Society* 2 (2), pp. 217–232.
- Enigbokan, A. (2011). Foolish journeys documentation. Archiving the City. [Online]. Retrieved on April 2011 from: <http://archivingthecity.com/2011/04/23/foolish-journeys-documentation/>.
- Foster, K. and Lorimer, H. (2007). Cultural geographies in practice: some reflections on art-geography as collaboration. *Cultural Geographies* 2007 14 (3), pp. 425–432.
- Gabrys, J. and Yusoff, K. (2012). Arts, sciences and climate change: practices and politics at the threshold. *Science as Culture* 21 (1), pp. 1–24.
- Galison, P. (1987). *How experiments end*. Chicago: University of Chicago Press.
- Gallagher, M. and Prior, J. (2010). Experimental research network – creative practice primer. [Online]. Retrieved on June 2011 from: <http://experimentalnetwork.org>.
- Garrett, B. (2011). Cracking the Paris carrières: corporal terror and illicit encounter under the City of Light. *ACME: An International E-Journal for Critical Geographies* 10 (2), pp. 269–277.
- Greenhough, B. (2010). Vitalist geographies. In: Anderson, B. and Harrison, P. (eds) *Taking place: non-representational theories and geography*. London: Ashgate, pp. 37–54.
- Gross, M. (2010). *Ignorance and surprise - science, society, and ecological design*. Cambridge, Mass: London MIT Press.
- Gross, M. and Hoffmann-Riem, H. (2005). Ecological restoration as a real-world experiment: designing robust implementation strategies in an urban environment. *Public Understanding of Science* 14, pp. 269–284.
- Gross, M. and Krohn, W. (2005). Society as experiment: sociological foundations for a self-experimental society. *History of the Human Sciences* 18 (2), pp. 63–86.
- Hacking, I. (1983). *Representing and intervening: introductory topics in the philosophy of natural science*. Cambridge: Cambridge University Press.
- Haraway, D. J. (2008). *When species meet*. Minnesota: University of Minnesota Press.
- Harris, L. and Hazen, H. (2006). Power of maps: (counter) mapping for conservation. *ACME: An International E-Journal for Critical Geographies* 4 (1), pp. 99–130.
- Harrison, S., Pile, S. and Thrift, N. (2004). *Patterned ground: entanglements of nature and culture*. London: Reaktion Books.
- Harvey, D. (1969). *Explanation in geography*. London: Arnold, Edward.
- Hawkins, H. (2011). Dialogues and doings: sketching the relationships between geography and art. *Geography Compass* 5 (7), pp. 464–478.
- Hemment, D., Ellis, R. and Wynne, B. (2011). Participatory mass observation and citizen science. *Leonardo Transactions* 44 (1), pp. 62–63.
- Hinchliffe, S., Kearnes, M. B., Degen, M. and Whatmore, S. (2005). Urban wild things: a cosmopolitical experiment. *Environment and Planning D: Society and Space* 23 (5), pp. 643–658.

- Hitchings, R. (2012). People can talk about their practices. *Area* 44 (1), pp. 61–67.
- Jasanoff, S. (2006). Experiments without borders: biology in the labs of life. LSE Public Lecture. London School of Economics, London. [Online]. Retrieved on May 2011 from: http://www2.lse.ac.uk/publicEvents/pdf/20060615_Jasanoff.pdf.
- Jellis, T. (2011). Geographer-artists: artist-geographers. Presented at the Association of American Geographers Annual Meeting, 15 April 2011. Seattle, USA.
- Kearnes, M. (2007). (Re)making matter: design and selection. *Area* 39 (2), pp. 143–155.
- Kearnes, M., Macnaghten, P. and Wilsdon, J. (eds) (2006). *Governing at the nanoscale: people, policies and emerging technologies*. London: Demos.
- Kearnes, M. and Wynne, B. (2007). On nanotechnology and ambivalence: the politics of enthusiasm. *Nanoethics* 1 (2), pp. 131–142.
- Kerr, I. (2008). Research and development: participatory research: geography: experimental. In: Thompson, N. (ed.) *Experimental geography*. New York: Melville House, pp. 63–77.
- Kitchin, R., Perkins, C. and Dodge, M. (2009). Thinking about maps. In: Dodge, M., Kitchin, R. and Perkins, C. (eds) *Rethinking maps*. London: Routledge, pp. 1–25.
- Krohn, W. and Weyer, J. (1994). Society as a laboratory: the social risks of experimental research. *Science and Public Policy* 21 (3), pp. 173–183.
- Krupar, S. R. (2007). Where eagles dare: an ethno-fable with personal land-fill. *Environment and Planning D* 25 (2), pp. 194–212.
- Krygier, J. and Wood, D. (2009). Ce n'est pas le monde'. In: Dodge, M., Kitchin, R. and Perkins, C. (eds) *Rethinking maps*. London: Routledge, pp. 189–218.
- Lancaster University. (2010). International Conference – The Experimental Society. Experimentality. Retrieved April 2011, from <http://www.lancs.ac.uk/experimentality/event/international-conference-experimental-society>.
- Lane, S., et al. (2011). Doing flood risk science differently: an experiment in radical scientific method. *Transactions of the Institute of British Geographers* 36 (1), pp. 15–36.
- Last, A. (2012a). Mutable matter – exploring nanotechnological futures through sensory methods. *Leonardo* 45 (2), pp. 132–139.
- Last, A. (2012b). Mutation – mapping in progress. In: Tyszczyk, R., Smith, J., Clark, N. and Butcher, M. (eds) *ATLAS: geography, architecture and change in an interdependent world*. London: Black Dog Publishing, pp. 162–163.
- Latham, A. (2003). Research, performance, and doing human geography: some reflections on the diary-photograph, diary-interview method. *Environment and Planning A* 35 (11), pp. 1993–2017.
- Latour, B. (2000). When things strike back: a possible contribution of 'science studies' to the social sciences. *British Journal of Sociology* 51 (1), pp. 107–123.
- Latour, B. (2004). Why has critique run out of steam? From matters of fact to matters of concern. *Critical Inquiry* 30 (2), pp. 225–248.
- Latour, B. and Woolgar, S. (1979). *Laboratory life - the social construction of scientific facts*. London: Sage.
- Law, J. (2004). *After method: mess in social science research*. London: Routledge.
- Loftus, A. (2012). *Everyday environmentalism: creating an urban political ecology*. Minneapolis: University of Minnesota Press.
- Lorimer, H. (2005). Cultural geography: the busyness of being 'more-than-representational'. *Progress in Human Geography* 29 (1), pp. 83–94.
- Lorimer, H. (2007). Cultural geography: worldly shapes, differently arranged. *Progress in Human Geography* 31 (1), pp. 89–100.
- Lorimer, H. (2008a). Cultural geography: non-representational conditions and concerns. *Progress in Human Geography* 32 (4), pp. 551–559.
- Lorimer, H. (2008b). Poetry and place: the shape of words. *Geography* 93 (3), pp. 181–182.
- Lorimer, H. and Wylie, J. (2010). LOOP (a geography). *Performance Research* 15 (4), pp. 6–13.
- Lury, C. and Wakeford, C. (2012). *Inventive methods: the happening of the social*. London: Routledge.
- Macpherson, H. M. (2005). Landscape's ocular-centrism – and beyond? In: Tress, B., Tress, G., Fry, G. and Opdam, P. (eds) *From landscape research to landscape planning: aspects of integration, education and application*. Wageningen: Wageningen UR Frontis Series, Springer; Kluwer Academic, pp. 95–104.
- Mann, A., et al. (2011). Mixing methods, tasting fingers notes on an ethnographic experiment. *HAU: Journal of Ethnographic Theory* 1 (1), pp. 221–243.
- Marres, N. (2009). Testing powers of engagement: green living experiments, the ontological turn and the undoability of involvement. *European Journal of Social Theory* 2009 12 (1), pp. 117–133.
- Massey, D. (1997). Spatial disruptions. In: Golding, S. (ed.) *The eight technologies of otherness*. London: Psychology Press/Routledge, pp. 208–214.
- Massey, D. (2005). *For space*. London: Sage.
- Massey, D. (2008). When theory meets politics. *Antipode* 40 (3), pp. 492–497.
- Massey, D. (2012a). *Talk given at the new maps for an island planet event at the southbank centre*. London. 13 March 2012. [online recording forthcoming]

- Massey, D. (2012b). Learning from Latin America. *Soundings* 50 (April 2012), pp. 131–141.
- McCormack, D. P. (2003). An event of geographical ethics in spaces of affect. *Transactions of the Institute of British Geographers* 28 (4), pp. 488–507.
- McCormack, D. P. (2010a). Thinking in transition: the affirmative refrain of experience/experiment. In: Anderson, B. and Harrison, P. (eds) *Taking place: non-representational theories and geography*. London: Ashgate, pp. 201–220.
- Merchant, S. (2011). The body and the senses: visual methods, videography and the submarine sensorium. *Body & Society* 17 (1), pp. 53–72.
- Paglen, T. (2008). Experimental geography: from cultural production to the production of space. In: Thompson, N. (ed.) *Experimental geography*. New York: Melville House, pp. 27–33.
- Patchett, M. (2011). Becoming geographer-artist. *Experimental Geography in Practice*. [Online]. Retrieved on June 2011 from: <https://merlepatchett.wordpress.com/2011/04/14/becoming-geographer-artist/>.
- Philips, R. (2006). *Sex, politics and empire: a postcolonial geography*. Manchester: Manchester University Press.
- Pile, S. and Thrift, N. (eds) (2000). *City A–Z: urban fragments*. London: Routledge.
- Pinder, D. (2005). Arts of urban exploration. *Cultural Geographies* 12 (4), pp. 383–411.
- Porteous, J. D. (1990). *Landscapes of the mind*. Toronto: University of Toronto Press.
- Powell, R. C. and Vasudevan, A. (2007). Geographies of experiment. *Environment and Planning A* 39 (8), pp. 1790–1793.
- Pratt, G. (2000). Research performances. *Environment and Planning D* 18 (5), pp. 639–651.
- Pratt, G. and Johnson, C. (2009). Translating research into theatre: nanay: a testimonial play. *B.C. Studies* 163, pp. 123–132.
- Ramsden, H. (2011). Walking & civic dialogue – a methodology for creative practice. Paper presented at the Association of American Geographers Annual Meeting, 15 April 2011. Seattle, USA.
- Rheinberger, H.-J. (1992). Experiment, difference, and writing: I. Tracing protein synthesis. *Studies in History and Philosophy of Science Part A* 23 (2), pp. 305–331.
- Rheinberger, H.-J. (2003). Scripts and scribbles. *Modern Language Notes* 118 (3), pp. 622–636.
- Rogoff, I. (2000). *Terra infirma: geography's visual culture*. London: Routledge.
- Ronell, A. (2005). *The test drive*. Urbana and Chicago: University of Illinois Press.
- Ronell, A. (2003). The experimental disposition: nietzsche's discovery of america (or, why the present administration sees everything in terms of a test). *American Literary History* 15 (3), pp. 560–574.
- Rosa, B., Garrett, B. L. and Prior, J. (2010). Creative practice primer project: Jute. Experimental Research Network. [Online]. Retrieved on June 2011 from: <http://experimentalnetwork.org/creative-practice-primer-2/creative-practice-primer-project-jute/>.
- Rose, G. (2003). On the need to ask how, exactly, is geography “Visual”? *Antipode* 35 (2), pp. 212–221.
- Salmelin, B. (2011). Participative innovation. Paper presented at Futur En Seine 2011. Paris. [Video URL: http://www.dailymotion.com/video/xjyb2x_bror-salmelin-le-futur-de-la-creation_tech]
- Shapin, S. and Schaffner, S. (1985). *Leviathan and the air-pump: hobbes, boyle and the experimental life*. Princeton: Princeton University Press.
- Stafford, R. A. (1989). *Scientist of Empire: Sir Robert Murchison, Scientific Exploration and Victorian Imperialism*. Cambridge: Cambridge University Press.
- Stengers, I. (1997). *Power and invention: situating science*. Minneapolis and London: University of Minnesota Press.
- Strathern, M. (2006). A community of critics? Thoughts on new knowledge. *Journal of the Royal Anthropological Institute* 12 (1), pp. 191–209.
- Szerszynski, B. (2005). Beating the unbound: political theatre in the laboratory without walls. In: Giannachi, G. and Stewart, N. (eds) *Performing nature: explorations in ecology and the arts*. Frankfurt and New York: Peter Lang, pp. 181–197.
- Thomke, S. H. (2003). *Experimentation matters: unlocking the potential of new technologies for innovation*. Boston, MA: Harvard Business School Press.
- Thompson, N. (ed.) (2008). *Experimental geography*. New York: Melville House.
- Thrift, N. (1991). Over-wordy worlds? In: Philo, C. (ed.) *New words, new worlds: reconceptualising social and cultural geography*. Lampeter: St David's University College, pp. 144–148.
- Thrift, N. (2000a). Dead or alive? In: Cook, I., Crouch, D., Naylor, S. and Ryan, J. (eds) *Cultural turns/geographical turns: perspectives on cultural geography*. Harlow, Essex: Prentice-Hall, pp. 1–6.
- Thrift, N. (2004). Summoning life. In: Cloke, P., Crang, P. and Goodwin, M. (eds) *Envisioning human geographies*. London: Arnold, pp. 81–103.
- Thrift, N. and Dewsbury, J. D. (2000). Dead geographies – and how to make them live. *Environment and Planning D* 18 (4), pp. 411–432.
- Tilley, H. (2011). *Africa as a living laboratory: empire, development, and the problem of scientific knowledge, 1870–1950*. Chicago: University of Chicago Press.
- Tolia-Kelly, D. P. (2010). *Landscape, race and memory: material ecologies of citizenship*. Farnham: Ashgate.
- Tuan, Y. F. (1977). *Space and place: the perspective of experience*. Minneapolis: University of Minnesota Press.

- White, N. (2010). Experimental fieldwork. Presented at *Experimentality*, Lancaster University. [<http://www.lancs.ac.uk/experimentality/blog/workshop-3-neal-white-experimentality-experimental-site>]
- Williamson, D. and Connolly, E. (2009). *Theirwork* – The development of sustainable mapping. In: Dodge, M., Kitchin, R. and Perkins, C. (eds) *Rethinking maps*. London: Routledge, pp. 94–112.
- Woodyer, T. (2012). Ludic geographies: not merely child's play. *Geography Compass* 6 (6), pp. 313–326.
- Wylie, J. (2010). Non-representational subjects? In: Anderson, B. and Harrison, P. (eds) *Taking place: non-representational theories and geography*. London: Ashgate, pp. 99–114.
- Yusoff, K. (2007). Antarctic exposure: archives of the feeling body. *Cultural Geographies* 14 (2), pp. 211–233.